Name____________________

ASSIGNMENT N.4 – TMM – A.Y. 2025-26

Closed book, closed notes, no help from other students nor from AI.
Show all work for full credit.
All justifications should use TMM basic principles.

For Questions 1- 4, match ALL correct answers a)-n) with the statements (answers can be used more than once and multiple answers are possible).

D) Ductile CI   G) Gray CI   M) Malleable CI   W) White CI   N) None of these

____ 1. Carbon in the form of Graphite spheres or clumps
____ 2. Brittle microstructure (typical ductility <2%)
____ 3. Formed by ladle additions of Mg
____ 4. Vibration damping capability leads to machine tool base applications

For Questions 5-9 match ALL correct answers a) – n) with the statements.

A) Austenitic S/S   F) Ferritic S/S   M) Martensitic
P) Recipitation hardenable   N) None of these

_____5. Heat treatable stainless steel
_____6. Has FCC crystal structure
_____7. Martensite is the primary microconstituent
_____8. Good choice for surgical knives .
_____9. Nonmagnetic

10. Suppose we plan to weld two pieces of a 6061-T6 (solution treated, quenched, and aged). We would be most concerned about what changes in heat affected zone, (HAZ)?

    a) formation hard brittle phases due to rapid cooling in the weld
    b) a decrease in strength due to overaging (formation of incoherent precipitate)
    c) a decrease in strength due to smaller grain sizes
    d) development of amorphous or glassy regions

11. Suppose we plan to weld two pieces of 4340 Quenched and Tempered steel. We would be most concerned about what changes in heat affected zone, (HAZ)?

    a) formation hard brittle phases due to rapid cooling in the weld
    b) a decrease in strength due to overaging (formation of incoherent precipitate)
    c) a decrease in strength due to smaller grain sizes
    d) development of amorphous or glassy regions

12. For a stainless steel to be stainless, it must contain a particular alloying element at a minimum concentration. Name the
    Alloying Element ____________   Minimum Concentration_____________
13. What is the function of the alloying element that makes stainless steels "stainless"

a. The element in the stainless steel acts as a solid solution strengthener.
b. Iron and the element form a second phase which provides good precipitation hardening.
c. The element reacts with oxygen to form a tenacious oxide layer on the surface, thereby preventing further oxidation.
d. The element increases the hardenability, which gives good hardness throughout the thickness.
e. None of the above.

In following Questions 14-18, for each type of metal listed

- Name one application for which it is particularly suited
- Justify the application by naming one or **four** attributes that would fulfill the design criteria for that application

Example:
Nickel Alloys
Application: Racing turbocharger rotor
Attribute: High temperature resistance means part is strong enough to operate at engine exhaust temps. Corrosion resistance means that part will retain geometry over time, etc.

14. Duplex steels
Application:
Attribute:

15. AA2XXX-T6
Application:
Attribute:

16. PH steels
Application:
Attribute:

17. Martensitic Stainless Steel
Application:
Attribute:

18. Gray cast iron
Application:
Attribute: